Disadvantages

1. Arranging the parts is very difficult.
2. It is more expensive.

4.10.3 Hypoid gears

In this type of gears the teeth are cut in spiral shape. In this arrangement the pinion gear is placed just below the center of the crown gear and hence the height of the chassis is reduced. Figure 4.10.3 shows the view of a hypoid gear arrangement.



Figure 4.10.3 Hypoid bevel gear

This type is mostly used in heavy vehicles. During its operation more heat is generated in it. Hence for reducing the heat generated, more amount of lubricating oil is required. It functions smoothly and works for long time.

Advantages

1. Safe to operate and it is less noisy in operation.
2. The height of chassis is less.
3. It can withstand maximum amount of rotational torque.

Disadvantages

1. It has low ground clearance.
2. Assembling and dismantling the parts is difficult. Repairing the parts is difficult.
3. It requires lubricating oil which must withstand high pressure.

Application

1. It is used mostly in the vehicles which are used for aged persons and patients.
2. It is used mostly in foreign cars.

4.10.4 Worm and worm wheel

In this type of gear in worm and worm wheel are used instead of pinion and crown wheel as shown in Figure 4.10.4. The worm in this arrangement is located as per the design either above or below the Worm Wheel. It is used mainly in heavy vehicles. It gives a smooth, superior and stable motion of operation. Worm is made of nickel and stabilized steel. Worm wheel is made up off phosphorous bronze material.



Figure 4.10.4 Worm and worm wheel arrangement

Advantages

1. Reduces the speed very quickly.
2. Transmission power is high.
3. Noiseless in operation.
4. By placing the worm at the upper side the ground clearance is increased.

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